

Dietary intake and the risk of coronary heart disease among the coconut-consuming Minangkabau in West Sumatra, Indonesia.

Several nutrition and non-nutritional pathways are recognised in the development and occurrence of cardiovascular disease. In many populations, high intakes of saturated fat are associated with elevated serum cholesterol concentrations and increased coronary heart disease (CHD) mortality. However, several studies report that hyperlipidaemia and heart diseases are not common among populations who consume coconut, a source of saturated fat. A case-control study was conducted among the Minangkabau known to be high coconut consumers to examine the difference in food patterns and risk of coronary heart disease (CHD) between the coronary cases and their gender- and age-matched apparently healthy counterparts serving as controls. Eligible subjects with CHD were identified through the co-operation of five participating hospitals located in Padang and Bukittinggi in West Sumatra, Indonesia. A total of 93 eligible cases (62 men and 31 women) in the Case group and 189 subjects (113 men and 76 women) in the Control group were recruited. Information on the intakes of individual foods and dishes over the preceding 12 months was obtained using a semi-quantitative food frequency questionnaire. The Case groups had significantly higher intakes of meats, eggs, sugar, tea, coffee and fruits, but lower intakes of soy products, rice and cereals compared to the controls. Coconut consumption as flesh or milk was not different between cases and controls. The cases had significantly higher intakes of protein and cholesterol, but lower intake of carbohydrate. Similar intakes of saturated and unsaturated fatty acids between the cases and controls indicated that the consumption of total fat or saturated fat, including that from coconut, was not a predictor for CHD in this food culture. However, the intakes of animal foods, total protein, dietary cholesterol and less plant derived carbohydrates were predictors of CHD.